

PS-02 — your data model

StayFinder — Hotel Discovery & Booking

Kognivera Hackathon 2026 · Travel & Tourism · data model v1.1.0-rc1

The problem statement itself, the 24-hour MVP scope and the XR device requirement live in the hackathon application, on your statement's page. This document is the data you have been given to build it with: every table, every field, and what each one is for.

You have **49,985 rows across 21 tables**. 18 of them are the tables this statement is built on; the remaining 3 are reference tables the others point at, included so the database works on its own.

All of it is in the `data/` folder beside this document: as `PS-02.db` (SQLite, indexed, ready to query), as CSV, and as DDL for Postgres and SQLite.

What the data gives you

300 hotels with **25 reviews each in eight languages** — the single ratio that decides whether your pros-and-cons summary is a feature or a parlour trick — plus 90 days of dated availability and a 100-case natural-language search evaluation set.

Watch out for this one

`hotel_reviews.language` is a BCP-47 tag and the bodies really are in that script. A summariser that assumes English will produce nonsense on about 58% of the corpus.

The tables this statement is built on

TABLE	ROWS	WHAT YOU USE IT FOR
<code>bookings</code>	1,996	The order header — where inventory, payment and itinerary meet. Idempotency key is mandatory, not optional.
<code>cities</code>	60	The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.
<code>countries</code>	30	ISO country reference. Every city, currency default and calling code resolves here.
<code>currencies</code>	25	carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always DECIMAL(12,2).
<code>hotel_room_types</code>	1,200	The bookable unit. APS-05's no-oversell guarantee is defended at this grain.
<code>hotels</code>	300	Fewer, richer properties. Depth (reviews, room types, media) matters more than catalogue size —.

TABLE	ROWS	WHAT YOU USE IT FOR
inventory_calendar	15,030	Entity x date availability with a database-level no-oversell CHECK. Deliberately scarce units so APS-05's load test has something to defend.
users	1,200	The traveller identity every personalisation hangs off. Segmented heavy / light / cold_start so APS-04 can prove cold start.
amenities	60	Replaces the pipe-separated amenity string. Filters in PS-02 join here.
eval_nl_search_set	100	100 natural-language hotel searches paired with the structured filters a correct parser should produce. Makes PS-02's NL search testable in seconds.
eval_queries	120	120 shared natural-language queries. Without them we get thirteen incomparable precision@k numbers.
hotel_amenities	2,400	Join table to the amenity master — the thing PS-02's amenity filter actually queries.
hotel_media	1,500	Image references with role and alt text. PS-05 uses hero/room rows as the fallback when no XR scene exists.
hotel_policies	300	Grounding material for PS-02's AI concierge — 'is it good for a family with a toddler?' is answered from here.
hotel_rate_plans	2,400	Makes rate selection a real choice rather than a single price.
hotel_reviews	7,500	the single most important density ratio in the pack. ~25 per hotel, mixed languages and traveller types, because PS-02's flagship feature is summarising them into balanced pros and cons.
payments	1,996	Mock payment records — authorised and captured amounts held separately so a partial capture is representable.
user_interactions	12,339	heavy users with long histories plus a deliberate cold-start cohort with none. One event per user makes everyone cold and APS-04 undemonstrable.

Reference tables, included so the database is valid

You will mostly join through these rather than think about them.

TABLE	ROWS	WHAT IT IS
itineraries	803	A versioned plan belonging to a trip. version is what makes PS-11's conflict handling tractable.
languages	26	Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.
trips	600	The container that gives dates, party, destination and budget to everything else. Seven statements produce or consume one.

How they fit together

Open `02_DATA_MODEL_DIAGRAM.html` in a browser for the clickable version — it shows these tables and nothing else. Download it first; it will not render inside SharePoint.

Some tables point at "any bookable thing" using an `(entity_type, entity_id)` pair rather than a typed foreign key. That is deliberate: it is what lets one feature refer to a hotel, a flight, a point of interest or a package without a separate join table for each. The legal values of `entity_type` are in `data/enums.json`.

Every field, table by table

Columns marked **PK** are the primary key. **FK** shows what a column points at. Enum columns list their legal values — anything else is rejected by the conformance check.

amenities

Reference & geography · 60 rows · IDs start `amn_`

Replaces the pipe-separated amenity string. Filters in PS-02 join here.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>amenity_id</code>	text	PK	amn_ prefixed.
<code>code</code>	text	UNIQUE · NOT NULL	snake_case machine code, e.g. free_wifi.
<code>label</code>	text	NOT NULL	Display label.
<code>amenity_group</code>	text	NOT NULL · one of <code>connectivity</code> , <code>wellness</code> , <code>food_beverage</code> , <code>family</code> , <code>accessibility</code> , <code>transport</code> , <code>business</code> , <code>outdoor</code>	
<code>icon_hint</code>	text		Suggested icon name, purely advisory.
<code>updated_at</code>	timestampz	NOT NULL	

currencies

Reference & geography · 25 rows · IDs start `cur_`

carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always DECIMAL(12,2).

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>currency_id</code>	text	PK	cur_ prefixed.
<code>iso4217</code>	char(3)	UNIQUE · NOT NULL	e.g. INR.
<code>name</code>	text	NOT NULL	
<code>symbol</code>	text	NOT NULL	
<code>minor_unit_exponent</code>	smallint	NOT NULL	0 for JPY/KRW, 2 default, 3 for KWD/BHD.

COLUMN	TYPE	CONSTRAINTS	NOTES
display_locale	text	NOT NULL	BCP-47 locale used for formatting.
updated_at	timestampz	NOT NULL	

inventory_calendar

Availability & pricing · 15,030 rows · IDs start `inv_`

Entity x date availability with a database-level no-oversell CHECK. Deliberately scarce units so APS-05's load test has something to defend.

COLUMN	TYPE	CONSTRAINTS	NOTES
inventory_id	text	PK	inv_ prefixed.
entity_type	text	NOT NULL · one of <code>room_type</code> , <code>flight_fare</code> , <code>guide</code> , <code>poi</code>	
entity_id	text	NOT NULL	room_type_id or fare_id.
for_date	date	NOT NULL	
total_units	int	NOT NULL	
booked_units	int	NOT NULL	
held_units	int	NOT NULL	
price	decimal(12,2)	NOT NULL	the price for that date.
currency	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	
min_stay_nights	smallint	NOT NULL	
closed_to_arrival	bool	NOT NULL	
updated_at	timestampz	NOT NULL	

Unique together: (`entity_type`, `entity_id`, `for_date`)

Enforced by the database: `booked_units + held_units <= total_units`

languages

Reference & geography · 26 rows · IDs start `lng_` · reference table

Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.

COLUMN	TYPE	CONSTRAINTS	NOTES
language_id	text	PK	lng_ prefixed.
bcp47	text	UNIQUE · NOT NULL	e.g. ta, hi, en-IN — never 'Tamil'.

COLUMN	TYPE	CONSTRAINTS	NOTES
english_name	text	NOT NULL	
native_name	text	NOT NULL	
script	text	NOT NULL	ISO-15924, e.g. Taml, Deva, Latn.
rtl	bool	NOT NULL	Right-to-left rendering flag.
tts_supported	bool	NOT NULL	Relevant to PS-13 voice output.
updated_at	timestampz	NOT NULL	

countries

Reference & geography · 30 rows · IDs start `cnt_`

ISO country reference. Every city, currency default and calling code resolves here.

COLUMN	TYPE	CONSTRAINTS	NOTES
country_id	text	PK	Canonical ID, cnt_ prefixed.
iso2	char(2)	UNIQUE · NOT NULL	ISO-3166-1 alpha-2, e.g. IN.
iso3	char(3)	UNIQUE · NOT NULL	ISO-3166-1 alpha-3, e.g. IND.
name	text	NOT NULL	English short name.
default_currency	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	ISO-4217 code.
calling_code	text	NOT NULL	E.164 country calling code, e.g. +91.
region	text	NOT NULL	UN sub-region grouping.
updated_at	timestampz	NOT NULL	Rule R4: UTC, ISO-8601 with offset.

cities

Reference & geography · 60 rows · IDs start `cty_`

The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.

COLUMN	TYPE	CONSTRAINTS	NOTES
city_id	text	PK	cty_ prefixed.
name	text	NOT NULL	City name.
state	text		State / province, nullable for city-states.
country_id	text	FK → <code>countries.country_id</code> · NOT NULL	

COLUMN	TYPE	CONSTRAINTS	NOTES
country_code	char(2)	NOT NULL	Denormalised ISO2 for convenient joins.
lat	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
lng	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
timezone	text	NOT NULL	IANA zone, e.g. Asia/Kolkata.
region	text	NOT NULL	Domestic region grouping, e.g. South India.
population	int		Approximate, for demand weighting.
season_profile	text	NOT NULL · one of winter, summer, monsoon, post_monsoon, spring, autumn	Dominant season at the peak travel window.
peak_months	text	NOT NULL	Comma-separated month numbers, e.g. 10,11,12.
primary_language	text	FK → languages.bcp47 · NOT NULL	Rule R6: BCP-47 tag.
description	text		One-paragraph orientation blurb, used by PS-13.
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

eval_nl_search_set

Signals & evaluation · 100 rows · IDs start ens_

100 natural-language hotel searches paired with the structured filters a correct parser should produce. Makes PS-02's NL search testable in seconds.

COLUMN	TYPE	CONSTRAINTS	NOTES
case_id	text	PK	ens_ prefixed.
utterance	text	NOT NULL	e.g. quiet 4-star near the beach with a pool under 8000.
language	text	FK → languages.bcp47 · NOT NULL	
expected_city_id	text	FK → cities.city_id	
expected_filters_json	text	NOT NULL	JSON string: star_min, price_max, amenities[], distance_km, etc.

COLUMN	TYPE	CONSTRAINTS	NOTES
expected_sort	text	NOT NULL	price_asc
difficulty	text	NOT NULL	easy
has_negation	bool	NOT NULL	e.g. 'not near the airport' — the cases parsers usually fail.
updated_at	timestampz	NOT NULL	

hotels

Supply & catalogue · 300 rows · IDs start `htl_`

Fewer, richer properties. Depth (reviews, room types, media) matters more than catalogue size —.

COLUMN	TYPE	CONSTRAINTS	NOTES
hotel_id	text	PK	htl_ prefixed.
city_id	text	FK → cities.city_id · NOT NULL	
name	text	NOT NULL	~3% near-duplicate names injected deliberately.
property_type	text	NOT NULL · one of hotel, resort, homestay, hostel, apartment, boutique, heritage, guesthouse	
star_rating	smallint	NOT NULL	1–5, CHECK constrained.
guest_score	decimal(2,1)		0.0–10.0; null for a handful of new properties.
review_count	int	NOT NULL	Denormalised count, must agree with hotel_reviews.
address_line	text	NOT NULL	
lat	decimal(9,6)	NOT NULL	
lng	decimal(9,6)	NOT NULL	
distance_to_centre_km	decimal(6,2)	NOT NULL	PS-02 filter: distance to a landmark.
description	text	NOT NULL	Plausible prose, embeddable.
base_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
checkin_time	text	NOT NULL	Local HH:MM at the property.

COLUMN	TYPE	CONSTRAINTS	NOTES
checkout_time	text	NOT NULL	
chain_code	text		Null for independents.
has_xr_scene	bool	NOT NULL	PS-05 / APS-07 — does an immersive preview exist.
status	text	NOT NULL · one of active, inactive, archived, draft	
created_at	timestampz	NOT NULL	
updated_at	timestampz	NOT NULL	

users

Identity & preference · 1,200 rows · IDs start `usr_`

The traveller identity every personalisation hangs off. Segmented heavy / light / cold_start so APS-04 can prove cold start.

COLUMN	TYPE	CONSTRAINTS	NOTES
user_id	text	PK	usr_ prefixed.
display_name	text	NOT NULL	Synthetic — no real people (content policy).
email	text	UNIQUE · NOT NULL	Synthetic @example.invalid addresses only.
home_city_id	text	FK → cities.city_id · NOT NULL	
home_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
locale	text	FK → languages.bcp47 · NOT NULL	UI language, BCP-47.
budget_band	text	NOT NULL · one of shoestring, value, mid, premium, luxury	
travel_style	text	NOT NULL · one of budget, comfort, luxury, adventure, slow, cultural, wellness	
traveller_type	text	NOT NULL · one of solo, couple, family, business, friends, senior, backpacker	
segment	text	NOT NULL · one of heavy, light, cold_start	heavy / light / cold_start cohorts.
date_of_signup	date	NOT NULL	
loyalty_tier	text		none

COLUMN	TYPE	CONSTRAINTS	NOTES
status	text	NOT NULL · one of active, inactive, archived, draft	
created_at	timestamptz	NOT NULL	
updated_at	timestamptz	NOT NULL	

eval_queries

Signals & evaluation · 120 rows · IDs start evq_

120 shared natural-language queries. Without them we get thirteen incomparable precision@k numbers.

COLUMN	TYPE	CONSTRAINTS	NOTES
query_id	text	PK	evq_ prefixed.
query_text	text	NOT NULL	
language	text	FK → languages.bcp47 · NOT NULL	Includes hi and ta queries for the multilingual requirement.
intent	text	NOT NULL · one of budget_stay, family_stay, luxury_stay, heritage_poi, nature_poi, food_poi, adventure_package, honeymoon_package, accessibility, pet_friendly	
target_entity_type	text	NOT NULL · one of hotel, room_type, rate_plan, flight, flight_fare, poi, package, package_component, guide, transfer, event, xr_scene	
city_id	text	FK → cities.city_id	Null for city-agnostic queries.
persona_user_id	text	FK → users.user_id	The user whose profile the query should be personalised to.
filters_json	text	NOT NULL	JSON string of the hard filters a correct system must apply.
k	smallint	NOT NULL	The k at which precision@k is reported.
notes	text		
updated_at	timestamptz	NOT NULL	

hotel_amenities

Supply & catalogue · 2,400 rows · IDs start `ham_`

Join table to the amenity master — the thing PS-02's amenity filter actually queries.

COLUMN	TYPE	CONSTRAINTS	NOTES
hotel_amenity_id	text	PK	ham_ prefixed.
hotel_id	text	FK → hotels.hotel_id · NOT NULL	
amenity_id	text	FK → amenities.amenity_id · NOT NULL	
is_free	bool	NOT NULL	
note	text		
updated_at	timestamptz	NOT NULL	

Unique together: (hotel_id, amenity_id)

hotel_policies

Supply & catalogue · 300 rows · IDs start `hpo_`

Grounding material for PS-02's AI concierge — 'is it good for a family with a toddler?' is answered from here.

COLUMN	TYPE	CONSTRAINTS	NOTES
policy_id	text	PK	hpo_ prefixed.
hotel_id	text	FK → hotels.hotel_id · UNIQUE · NOT NULL	
child_policy	text	NOT NULL	Prose.
pet_policy	text	NOT NULL	Prose.
extra_bed_policy	text	NOT NULL	
extra_bed_charge	decimal(12,2)		
extra_bed_currency	char(3)	FK → currencies.iso4217	
payment_methods	text	NOT NULL	Comma-separated payment_method enum values.
airport_pickup	bool	NOT NULL	Directly answers a scripted concierge question.
early_checkin_possible	bool	NOT NULL	
accessibility_notes	text		
updated_at	timestamptz	NOT NULL	

hotel_room_types

Supply & catalogue · 1,200 rows · IDs start `rmt_`

The bookable unit. APS-05's no-oversell guarantee is defended at this grain.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>room_type_id</code>	text	PK	<code>rmt_</code> prefixed.
<code>hotel_id</code>	text	FK → <code>hotels.hotel_id</code> · NOT NULL	
<code>name</code>	text	NOT NULL	e.g. Deluxe Garden View.
<code>max_occupancy</code>	smallint	NOT NULL	
<code>max_adults</code>	smallint	NOT NULL	
<code>max_children</code>	smallint	NOT NULL	
<code>bed_config</code>	text	NOT NULL · one of <code>single</code> , <code>twin</code> , <code>double</code> , <code>queen</code> , <code>king</code> , <code>bunk</code> , <code>twin_double</code>	
<code>size_sqm</code>	smallint		
<code>base_rate</code>	decimal(12,2)	NOT NULL	NUMERIC, never FLOAT.
<code>currency</code>	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	always paired with the amount.
<code>total_units</code>	int	NOT NULL	Deliberately scarce on some rows so APS-05 has contention.
<code>smoking_allowed</code>	bool	NOT NULL	
<code>status</code>	text	NOT NULL · one of <code>active</code> , <code>inactive</code> , <code>archived</code> , <code>draft</code>	
<code>updated_at</code>	timestampz	NOT NULL	

trips

Trip & itinerary · 600 rows · IDs start `trp_` · reference table

The container that gives dates, party, destination and budget to everything else. Seven statements produce or consume one.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>trip_id</code>	text	PK	<code>trp_</code> prefixed.
<code>owner_user_id</code>	text	FK → <code>users.user_id</code> · NOT NULL	
<code>title</code>	text	NOT NULL	
<code>origin_city_id</code>	text	FK → <code>cities.city_id</code>	

COLUMN	TYPE	CONSTRAINTS	NOTES
destination_city_id	text	FK → cities.city_id · NOT NULL	
start_date	date	NOT NULL	Rule R4: zoneless calendar date.
end_date	date	NOT NULL	
party_size	smallint	NOT NULL	
adults	smallint	NOT NULL	
children	smallint	NOT NULL	
trip_type	text	NOT NULL · one of solo, couple, family, business, friends, senior, backpacker	
is_group_trip	bool	NOT NULL	PS-11 / PS-08 filter.
status	text	NOT NULL · one of draft, planning, confirmed, in_progress, completed, cancelled	
home_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
notes	text		
created_at	timestamptz	NOT NULL	
updated_at	timestamptz	NOT NULL	

user_interactions

Signals & evaluation · 12,339 rows · IDs start uix_

heavy users with long histories plus a deliberate cold-start cohort with none. One event per user makes everyone cold and APS-04 undemonstrable.

COLUMN	TYPE	CONSTRAINTS	NOTES
interaction_id	text	PK	uix_ prefixed.
user_id	text	FK → users.user_id · NOT NULL	
entity_type	text	NOT NULL · one of hotel, room_type, rate_plan, flight, flight_fare, poi, package, package_component, guide, transfer, event, xr_scene	
entity_id	text	NOT NULL	
interaction_type	text	NOT NULL · one of view, click, like, save, book, dismiss, share, search	
occurred_at	timestamptz	NOT NULL	

COLUMN	TYPE	CONSTRAINTS	NOTES
dwelt_seconds	int		Null for non-view events.
position_in_list	smallint		Rank at which the item was shown — needed for unbiased offline eval.
query_text	text		Set for interaction_type = search.
query_language	text	FK → languages.bcp47	
channel	text	NOT NULL · one of web, mobile_app, partner, call_centre, agent	
session_id	text	NOT NULL	
implicit_rating	decimal(3,2)		Derived signal, provided for convenience.

hotel_media

Supply & catalogue · 1,500 rows · IDs start hmd_

Image references with role and alt text. PS-05 uses hero/room rows as the fallback when no XR scene exists.

COLUMN	TYPE	CONSTRAINTS	NOTES
media_id	text	PK	hmd_ prefixed.
hotel_id	text	FK → hotels.hotel_id · NOT NULL	
room_type_id	text	FK → hotel_room_types.room_type_id	Null for property-level media.
media_role	text	NOT NULL · one of hero, room, lobby, exterior, dining, pool, bathroom, view, landmark, menu, sign, receipt	
file_path	text	NOT NULL	Relative path inside the media pack.
alt_text	text	NOT NULL	Accessibility and multimodal grounding.
width_px	int	NOT NULL	
height_px	int	NOT NULL	
sort_order	smallint	NOT NULL	
updated_at	timestampz	NOT NULL	

hotel_rate_plans

Supply & catalogue · 2,400 rows · IDs start `rtp_`

Makes rate selection a real choice rather than a single price.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>rate_plan_id</code>	text	PK	<code>rtp_</code> prefixed.
<code>room_type_id</code>	text	FK → <code>hotel_room_types.room_type_id</code> · NOT NULL	
<code>plan_type</code>	text	NOT NULL · one of <code>refundable</code> , <code>non_refundable</code> , <code>breakfast_included</code> , <code>half_board</code> , <code>full_board</code> , <code>long_stay</code>	
<code>name</code>	text	NOT NULL	
<code>price_delta</code>	decimal(12,2)	NOT NULL	Signed delta applied to the room <code>base_rate</code> .
<code>currency</code>	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	
<code>cancellation_window_hours</code>	int	NOT NULL	0 for non-refundable.
<code>cancellation_penalty_pct</code>	smallint	NOT NULL	
<code>includes_breakfast</code>	bool	NOT NULL	
<code>min_stay_nights</code>	smallint	NOT NULL	
<code>status</code>	text	NOT NULL · one of <code>active</code> , <code>inactive</code> , <code>archived</code> , <code>draft</code>	
<code>updated_at</code>	timestampz	NOT NULL	

hotel_reviews

Supply & catalogue · 7,500 rows · IDs start `rvw_`

the single most important density ratio in the pack. ~25 per hotel, mixed languages and traveller types, because PS-02's flagship feature is summarising them into balanced pros and cons.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>review_id</code>	text	PK	<code>rvw_</code> prefixed.
<code>hotel_id</code>	text	FK → <code>hotels.hotel_id</code> · NOT NULL	
<code>user_id</code>	text	FK → <code>users.user_id</code>	Nullable — some reviews are from non-registered guests.
<code>rating</code>	smallint	NOT NULL	1–10.
<code>title</code>	text		

COLUMN	TYPE	CONSTRAINTS	NOTES
body	text	NOT NULL	Plausible prose, 40–120 words.
language	text	FK → languages.bcp47 · NOT NULL	Rule R6: mixed en-IN / hi / ta / ml / bn.
traveller_type	text	NOT NULL · one of solo, couple, family, business, friends, senior, backpacker	
stay_date	date	NOT NULL	Rule R4: zoneless.
room_type_id	text	FK → hotel_room_types.room_type_id	
helpful_votes	int	NOT NULL	
has_photo	bool	NOT NULL	
sentiment_hint	decimal(3,2)		-1.00..1.00, provided for calibration only.
created_at	timestamptz	NOT NULL	
updated_at	timestamptz	NOT NULL	

itineraries

Trip & itinerary · 803 rows · IDs start `itn_` · reference table

A versioned plan belonging to a trip. version is what makes PS-11's conflict handling tractable.

COLUMN	TYPE	CONSTRAINTS	NOTES
itinerary_id	text	PK	itn_ prefixed.
trip_id	text	FK → trips.trip_id · NOT NULL	
name	text	NOT NULL	
version	int	NOT NULL	Monotonic per itinerary.
is_active	bool	NOT NULL	Exactly one active version per trip.
generated_by	text	NOT NULL · one of user, ai_planner, optimizer, agent, vote, import	Which subsystem produced this version.
total_cost	decimal(12,2)	NOT NULL	sum of item costs, half-up at the end.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
total_duration_minutes	int	NOT NULL	
total_carbon_kg	decimal(10,3)	NOT NULL	APS-09 objective.

COLUMN	TYPE	CONSTRAINTS	NOTES
optimizer_weights	text		JSON string: {cost, time, carbon} weights that produced it.
status	text	NOT NULL · one of active, inactive, archived, draft	
created_at	timestampz	NOT NULL	
updated_at	timestampz	NOT NULL	

bookings

Booking & money · 1,996 rows · IDs start `bkg_`

The order header — where inventory, payment and itinerary meet. Idempotency key is mandatory, not optional.

COLUMN	TYPE	CONSTRAINTS	NOTES
booking_id	text	PK	bkg_ prefixed.
user_id	text	FK → users.user_id · NOT NULL	
trip_id	text	FK → trips.trip_id	Null for standalone bookings.
itinerary_id	text	FK → itineraries.itinerary_id	
booking_reference	text	UNIQUE · NOT NULL	Human-quotable 6-char reference.
channel	text	NOT NULL · one of web, mobile_app, partner, call_centre, agent	
total_amount	decimal(12,2)	NOT NULL	must equal the sum of booking_items.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
tax_amount	decimal(12,2)	NOT NULL	
idempotency_key	text	UNIQUE · NOT NULL	APS-05 — retried requests resolve to this same row.
status	text	NOT NULL · one of pending, confirmed, partially_confirmed, cancelled, failed, refunded	
confirmed_at	timestampz		
cancelled_at	timestampz		
cancellation_reason	text		
created_at	timestampz	NOT NULL	

COLUMN	TYPE	CONSTRAINTS	NOTES
updated_at	timestampz	NOT NULL	

payments

Booking & money · 1,996 rows · IDs start `pay_`

Mock payment records — authorised and captured amounts held separately so a partial capture is representable.

COLUMN	TYPE	CONSTRAINTS	NOTES
payment_id	text	PK	pay_ prefixed.
booking_id	text	FK → bookings.booking_id · NOT NULL	
method	text	NOT NULL · one of card, upi, netbanking, wallet, mock	
status	text	NOT NULL · one of initiated, authorised, captured, failed, refunded, voided	
authorised_amount	decimal(12,2)	NOT NULL	D3.
captured_amount	decimal(12,2)	NOT NULL	
refunded_amount	decimal(12,2)	NOT NULL	
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
gateway_reference	text	NOT NULL	Synthetic mock reference.
idempotency_key	text	UNIQUE · NOT NULL	
failure_code	text	one of hold_expired, sold_out, over_budget, invalid_id, currency_mismatch, idempotency_conflict, constraint_infeasible, low_confidence	
authorised_at	timestampz		
captured_at	timestampz		
created_at	timestampz	NOT NULL	
updated_at	timestampz	NOT NULL	

The rules that apply to these fields

#	RULE
R1	Additive only. Add columns, tables and stores freely. Never rename, drop or repurpose a field that came with the data.
R2	IDs are opaque prefixed strings — <code>htl_a91f3c</code> . Never integers, never parsed for meaning.
R3	Money is a pair: a 2-place decimal plus an ISO-4217 currency code. Never a float.
R4	Time is ISO-8601 with an offset. <code>_at</code> fields carry an offset; <code>_date</code> fields have no zone.
R5	Enums are lowercase snake_case and the legal values are in <code>data/enums.json</code> .
R6	Language is a BCP-47 tag — <code>ta</code> , not "Tamil".
R7	Geography is WGS-84 to 6 decimal places, <code>lat</code> and <code>lng</code> together or not at all.
R8	Nothing is hard-deleted. Rows carry <code>status</code> and <code>updated_at</code> .

Add whatever you like beside these fields — new columns, new tables, your own vector store, your own services. That is the point of R1. What you must not do is rename or re-key the fields that came with the data, because that is what would stop sixteen independent builds being put together afterwards.

`data/WORKING_WITH_THE_DATA.md` has the loading instructions, including how to read money without corrupting it. `tools/validate_conformance.py` tells you in thirty seconds whether you are still conformant.